

ENDOCRINE PHYSIOLOGY OBJECTIVES

ENDOCRINE PHYSIOLOGY OVERVIEW AND PRINCIPLES.

1. Know the major endocrine organs and the hormones they secrete.
2. What are the three major chemical classes of endocrine hormones?
3. Know the chemical class of all of the major hormones discussed.
4. Know the major steps in protein hormone and steroid hormone biosynthesis. What are the roles of the pre and pro segments of prehormones?
5. Understand the mechanism of storage and secretion of protein/peptide hormones from endocrine cells.
6. Know the principle mechanisms used for inactivating or metabolizing hormones following secretion and the clinical significance of liver and renal disease on these processes.
7. Know predominant mechanisms used to metabolize steroid vs. protein hormones
8. Be able to distinguish the general mechanisms used for hormone blood transport between steroid hormones vs. protein hormones and catecholamines.
9. What are the physiological purposes of steroid hormone binding by plasma proteins? What is the functional difference between a bound vs. free (dissolved) hormone? What impact does predominant vs minimal hormone binding have on hormone concentrations in the blood?
10. Know the two basic classes of endocrine hormone receptors on target cells and the subtypes within each class.
11. Be able to identify the basic receptor type(s) utilized by the three principle structural classes of hormones.
12. Know the two basic types of second messenger pathways used by G-protein receptors?
13. Which receptor type(s) predominantly regulate target cells via a phosphorylation cascade?
14. Which receptor type(s) predominantly regulate target cells via a transcription/translation mechanism?
15. Why is steroid and thyroid hormone action slower on target cells compared to protein hormones?
16. Know the basic class of membrane receptor used by all major protein/peptide hormones. Be able to identify the three hormones discussed that act via a tyrosine kinase receptor instead of a G-protein receptor.
17. Understand the concepts of one hormone affecting many different processes; one hormone affecting many different aspects a single process and multiple hormones affecting a single process.
18. Know the principle factors that can alter normal hormone receptor action on target cells and the clinical relevance. What is the major factor that affects tissue hormone sensitivity? Be able to predict how changes in tissue sensitivity affect tissue responsiveness to hormones and understand the clinical relevance.
19. Be able to describe the basic mechanism of negative feedback inhibition control of hormone secretion.
20. Know the four basic physiological processes regulated by endocrine hormones and understand the complexities of a single hormone regulating multiple processes and single processes regulated by multiple hormones.

21. Understand that endocrine disorders invariably result from either deficient or excess hormone secretion and/or action and know the basic clinical algorithm used for clinical diagnosis.
22. Understand why the non-uniform secretion pattern of most hormones complicates the clinical diagnosis of endocrine disorders.